

Author Contribution Report

Distributed Node Health Monitoring Framework

June 24, 2026

1 Project Information

Paper Title: Distributed Node Health Monitoring Framework
Course / Research Group: Distributed Systems
Supervisor: Dr. Tran Huu Tam
Submission Date: 24 Jun 2026

2 Authors

#	Name	Student ID / Affiliation
1	Nguyen Ha Khai	10423053
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3	Vo Minh Duy	10423028
4	Ninh Hoang Quan	10423097
5	To Huynh Phuc	10423093

3 Contribution Matrix (CRediT)

Contribution Role	NHK	NDK	VMD	NHQ	THP
Conceptualization	P	S	S	P	S
Methodology	S	P	S	–	P
Software Development	P	P	–	P	S
Validation	S	S	–	–	P
Formal Analysis	P	P	–	–	P
Data Collection / Experiments	–	–	–	–	P
Visualization	S	S	P	S	P
Writing	P	P	P	P	P
Project Administration	P	–	–	P	S
Literature Review	P	P	P	–	–

Use symbols such as:

- P = Primary Contributor
- S = Supporting Contributor
- – = No Contribution

4 Detailed Contributions

Author 1: Nguyen Ha Khai

Primary Roles

- Co-leader with Ninh Hoang Quan
- Algorithm analyst
- Manuscript Editor

Specific Contributions

- Implement FINDING NEMO & CHUALANH
- Write the FINDING NEMO component in implementation details and proposed approach section
- Write the system design part in related work section

Author 2: Nguyen Duy Khanh

Primary Roles

- Software engineer
- Algorithm designer

Specific Contributions

- Implement hybrid SWIMPHI
- Design CHUALANH algorithm
- Review and evaluation the practice of different failure detection approaches
- Write the hybrid SWIMPHI component in implementation details and proposed approach section
- Write the failure detection part in related work section

Author 3: Vo Minh Duy

Primary Roles

- Frontend developer
- Literature reviewer

Specific Contributions

- Design the monitoring dashboard for data collection and system performance evaluation
- Review 90% citations in the research paper
- Research the architecture of the system
- Write the abstract, introduction, problem statement, and complete architectural diagram

Author 4: Ninh Hoang Quan

Primary Roles

- Backend developer
- Leader

Specific Contributions

- Implement the backbone or foundation of nodes in the system
- Write the CHUALANH algorithm in implementation details and proposed approach section

Author 5: To Huynh Phuc

Primary Roles

- Manual tester
- Code reviewer
- Algorithm designer

Specific Contributions

- Design FINDING NEMO and hybrid SWIMPHI algorithm
- Write the evaluation; conclusion and future work sections
- Refactor the source code and ensure cohesive interactions between system components

5 Author Contribution Statement

The authors contributed to this work according to the CRediT (Contributor Roles Taxonomy) framework.

- **Author 1:** Conceptualization, Methodology, Software, Writing – Original Draft.
- **Author 2:** Validation, Investigation, Writing – Review & Editing.
- **Author 3:** Software, Visualization, Formal Analysis.
- **Author 4:** Literature Review, Investigation.
- **Author 5:** Project Administration, Validation.

All authors reviewed and approved the final manuscript.

6 Estimated Contribution Percentages (Optional)

Some institutions require percentage estimates in addition to the CRediT taxonomy.

Author	Contribution (%)
Author 1	20
Author 2	20
Author 3	20
Author 4	20
Author 5	20
Total	100%

7 Declaration

We certify that the contribution information provided in this report accurately reflects each author's participation in the project and manuscript preparation.

Author Signature

Date
